PINN_M3DC1

Module 2

Synthetic MHD Solver and Case Generation

A textbook-style numerical-physics specification for periodic magnetic-island coalescence

Module purpose

Generate reproducible, verified full numerical-field trajectories of a two-dimensional reduced resistive-MHD island-coalescence problem, together with spatial and temporal convergence evidence suitable as reference data for a full-field solution-operator surrogate.

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Physics and numerical-method note for the synthetic reduced-MHD proof of concept

Preface: what Module 2 must establish

Module 1 defined the physical model. Module 2 must now answer a different question: how do we turn those continuous partial differential equations into trustworthy numerical trajectories?

That question cannot be answered by producing an attractive animation. A numerical field can look smooth and plausible while containing a sign error, an aliased nonlinear term, a poorly resolved current sheet, an unstable time step, or a false reconnection event caused by numerical diffusion. Because the later surrogate learns from the complete numerical field arrays produced here, every systematic solver error becomes part of the learning target. Module 2 is therefore the scientific foundation beneath Modules 3–6.

The initials **MHD** mean *magnetohydrodynamics*. A **synthetic case** is a numerical trajectory generated by the reduced model selected by this project, rather than an experimental measurement or a genuine M3D-C1 calculation. A **pseudo-spectral method** represents periodic fields by Fourier modes, differentiates those modes algebraically, and evaluates nonlinear products on a physical grid. A **convergence study** demonstrates that selected numerical answers approach a stable limit when the spatial and temporal discretizations are refined. For the revised surrogate formulation, a **full numerical field** means the two-dimensional array of scalar values on the accepted physical grid. Those arrays, not contour plots, color images, streamlines, or screenshots, form the downstream learning interface.

Important distinction

The word *synthetic* does not mean arbitrary or unverified. It means that the data are generated by the stated two-field reduced-MHD initial-value problem. These data can support a reduced-MHD full-field surrogate proof of concept. They cannot, by themselves, support a claim of fidelity to M3D-C1. Formatting the arrays to resemble field-oriented output from a larger code does not make them M3D-C1 data.

How to read this note

The main dependency chain is

$$\begin{gathered}\text{frozen reduced-MHD equations} \longrightarrow \text{periodic initial-value problem} \longrightarrow \text{Fourier representation}\\ \longrightarrow \text{dealiased nonlinear evaluation} \longrightarrow \text{time integration} \longrightarrow \text{diagnostics}\\ \longrightarrow \text{verification and convergence} \longrightarrow \text{accepted HDF5 trajectory.}\end{gathered}$$

Sections 1–3 define the exact physical case. Sections 4–6 derive the numerical method. Sections 8–10 define the evidence required before a trajectory is accepted. Sections 11–13 freeze the case-generation and implementation contracts. The appendices collect derivations, symbols, terminology, and concept questions.

Learning objectives

After working through this module, the reader should be able to:

1. distinguish a mathematical solution, a numerical approximation, a stored trajectory, and a rendered plot;
2. state exactly which fields the solver advances and which fields it reconstructs;
3. derive the periodic island-coalescence initial fields and locate their O-points and X-points;
4. explain why the unperturbed magnetic array is an equilibrium of the ideal nonlinear equations;
5. explain how the stream-function perturbation drives two parallel current channels together;
6. represent a periodic field by discrete Fourier coefficients and differentiate it spectrally;
7. invert $\nabla_\perp^2 U = \omega$ while treating the zero Fourier mode correctly;
8. explain how a pseudo-spectral method differs from a fully spectral Galerkin method;
9. derive the Fourier-space right-hand sides for ψ and ω;
10. explain aliasing and apply the two-thirds truncation rule;
11. implement one classical fourth-order Runge–Kutta step without using stale derived fields;
12. monitor advective, Alfvénic, and diffusive time-step restrictions;
13. define energy, current-sheet, symmetry, topology, and reconnection diagnostics;
14. distinguish code verification from model validation;
15. conduct separate temporal and spatial refinement studies;
16. calculate observed convergence order and self-convergence error;
17. decide whether a current sheet is resolved using spectra and physical-space measures;
18. define a reproducible parameter sweep without accepting unconverged cases;
19. distinguish full numerical scalar-field arrays from rendered images and explain why only the former are suitable learning objects;
20. state the full-field solution-operator interface $(\psi_0, U_0, \eta, \nu, t) \mapsto (\psi_t, U_t)$;
21. explain why A_0 remains case metadata even though its value is already encoded in the frozen-family input field U_0;

22. explain why many time-conditioned field pairs from one trajectory do not create many independent physical cases;

23. identify the metadata required to reproduce and losslessly reconstruct a stored trajectory; and

24. state precisely what evidence Module 2 does and does not provide.

Contents

1 Scientific role and acceptance boundary

1.1 The numerical object being built

For a fixed parameter vector

$$\boldsymbol{\mu} = (\eta, \nu, A_0), \tag{1.1}$$

the solver approximates a trajectory

$$\mathcal{Q}_{\boldsymbol{\mu}} = \{\psi(x,y,t), U(x,y,t), J(x,y,t), \omega(x,y,t) : (x,y) \in \Omega,\ 0 \le t \le T\}. \tag{1.2}$$

Here η is magnetic diffusivity, ν is kinematic viscosity, and A_0 is the amplitude of the initial flow perturbation. All three are dimensionless under the physical normalization frozen in Module 1.

The mathematical state is continuous. A computed case is finite:

$$\mathcal{Q}_{\boldsymbol{\mu}}^{N_x,N_y,\Delta t} = \{\psi_{ijn}, U_{ijn}, J_{ijn}, \omega_{ijn}\}, \tag{1.3}$$

where i and j label grid points and n labels saved times. The superscripts emphasize that the stored result depends on resolution and time step. The convergence program must show that this dependence is acceptably small for the quantities used later.

For the revised downstream surrogate, the physical case descriptor $\boldsymbol{\mu}$ and the neural input are deliberately kept distinct. Module 2 still generates cases by choosing (η, ν, A_0), but the primary learning interface uses the complete initial fields:

$$\boxed{\mathcal{G} : (\psi_0(\cdot), U_0(\cdot), \eta, \nu, t) \longmapsto (\psi(\cdot, t), U(\cdot, t)).} \tag{1.4}$$

On the accepted grid, each ψ_0, U_0, ψ_t, and U_t is an $N_x \times N_y$ array of scalar numerical values. The notation "$\cdot$" means the whole spatial field, not one queried coordinate. Current J and vorticity ω remain stored verification and diagnostic fields and may be derived from ψ and U; they are not required as primary outputs of the version-1 operator surrogate.

For the frozen initial-condition family, A_0 determines the complete field $U_0 = A_0(\cos x - \cos y)$. Therefore a separate A_0 neural input is redundant once the full U_0 array is supplied. Module 2 nevertheless stores A_0 as physical case metadata for reproducibility, stratification, and later analysis.

1.2 Four levels of evidence

Level	Question	Required evidence
Definition	Is the intended problem unambiguous?	Equations, signs, initial conditions, periodicity, gauges, parameters, and diagnostics are frozen.
Verification	Does the code solve those equations correctly?	Exact modes, manufactured solutions, numerical-operator tests, balance laws, parity, and convergence.
Numerical adequacy	Is this particular nonlinear trajectory resolved?	Spectral tails, current-sheet resolution, stable diagnostics, and refinement agreement.
Validation	Does the reduced model represent a target physical system or M3D-C1?	External observations or genuine higher-fidelity calculations. This evidence is not produced by Module 2.

Definition

Verification asks whether the equations were solved correctly. **Validation** asks whether the chosen equations represent the physical target adequately. A code can be verified for a reduced model without that reduced model being validated as a substitute for M3D-C1.

1.3 Module 2 deliverables

Module 2 is complete only when the project can produce:

1. an independently reproducible Python trajectory;
2. an independently reproducible MATLAB trajectory using the same mathematical contract;
3. a baseline island-coalescence case with complete metadata;
4. an exact-mode or manufactured-solution verification report;
5. a temporal-refinement report;
6. a spatial-refinement and spectral-resolution report;
7. energy, mean, periodicity, divergence, symmetry, and topology histories;
8. Python–MATLAB parity metrics on a common saved grid;
9. lossless HDF5 full-field arrays that can be reconstructed identically in either language;
10. an operator-ready handoff view that maps each accepted initial field to selected later field snapshots without rendering; and

11. a case manifest that records whether every acceptance criterion passed.

Required result

A single contour plot is not a Module 2 deliverable and is never a learning datum. The accepted object is a numerical field trajectory plus enough evidence and metadata for another implementation to reproduce, audit, and losslessly construct the full-field operator pairs in Eq. (1.4).

2 Frozen reduced-MHD initial-value problem

2.1 Domain and periodicity

The version-1 domain is the half-open square

$$\Omega = [0, L_x) \times [0, L_y), \qquad L_x = L_y = 2\pi. \tag{2.1}$$

The half-open notation means that $x = 0$ is stored but $x = 2\pi$ is not stored; those positions are the same periodic point. The same convention applies in y.

For every field and every derivative needed by the equations,

$$f(x + 2\pi, y, t) = f(x, y, t), \tag{2.2}$$
$$f(x, y + 2\pi, t) = f(x, y, t). \tag{2.3}$$

Fourier representation enforces this smooth periodic structure by construction.

2.2 Fields and signs

The primary physical fields are the magnetic-flux function ψ and flow stream function U. The frozen Module 1 definitions are

$$\boldsymbol{B}_\perp = \nabla\psi \times \hat{\boldsymbol{z}} = (\psi_y, -\psi_x), \tag{2.4}$$
$$\boldsymbol{v}_\perp = \hat{\boldsymbol{z}} \times \nabla U = (-U_y, U_x), \tag{2.5}$$
$$J = -\nabla_\perp^2 \psi, \tag{2.6}$$
$$\omega = \nabla_\perp^2 U, \tag{2.7}$$
$$[f, g] = f_x g_y - f_y g_x. \tag{2.8}$$

Consequently,

$$[U, f] = \boldsymbol{v}_\perp \cdot \nabla f. \tag{2.9}$$

2.3 Evolution equations

The solver advances

$$\partial_t \psi + [U, \psi] = \eta \nabla_\perp^2 \psi, \tag{2.10}$$
$$\partial_t \omega + [U, \omega] = [J, \psi] + \nu \nabla_\perp^2 \omega, \tag{2.11}$$
$$J = -\nabla_\perp^2 \psi, \qquad \omega = \nabla_\perp^2 U. \tag{2.12}$$

Equivalently,

$$\partial_t \psi = -[U, \psi] - \eta J, \tag{2.13}$$
$$\partial_t \omega = -[U, \omega] + [J, \psi] + \nu \nabla_\perp^2 \omega. \tag{2.14}$$

The solver's most convenient prognostic variables are (ψ, ω). At every time-integration stage it reconstructs U from $\nabla_\perp^2 U = \omega$ and derives $J = -\nabla_\perp^2 \psi$. The stored dataset fields remain (ψ, U, J, ω).

Frozen Module 2 convention

The code must never advance four independent arrays for ψ, U, J, and ω. Doing so would permit violations of $J = -\nabla_\perp^2 \psi$ and $\omega = \nabla_\perp^2 U$. Advance (ψ, ω), reconstruct (U, J) at every stage, and store all four only after their definitions agree.

2.4 Gauge and zero-mode constraints

Only derivatives of U affect the physical velocity, so

$$\langle U \rangle (t) = 0 \tag{2.15}$$

fixes its arbitrary additive constant. The initial flux has zero mean, and the periodic equation preserves

$$\langle \psi \rangle (t) = 0. \tag{2.16}$$

Because J and ω are periodic Laplacians,

$$\langle J \rangle = \langle \omega \rangle = 0. \tag{2.17}$$

These are exact compatibility conditions. They should be enforced at the zero Fourier mode and monitored in physical space.

2.5 Total-energy identity

The magnetic, kinetic, and total energies are

$$E_B(t) = \frac{1}{2}\int_\Omega |\nabla\psi|^2 \, \mathrm{d}A, \tag{2.18}$$

$$E_K(t) = \frac{1}{2}\int_\Omega |\nabla U|^2 \, \mathrm{d}A, \tag{2.19}$$

$$E(t) = E_B(t) + E_K(t). \tag{2.20}$$

Smooth periodic solutions satisfy

$$\boxed{\frac{\mathrm{d}E}{\mathrm{d}t} = -\eta\int_\Omega J^2 \, \mathrm{d}A - \nu\int_\Omega \omega^2 \, \mathrm{d}A.} \tag{2.21}$$

This identity is the principal global sign and dissipation audit, but it is not sufficient by itself: many wrong fields can share a similar energy history.

3 The periodic island-coalescence benchmark

3.1 Why island coalescence is useful

Two parallel current channels attract magnetically. In a two-dimensional flux map, each channel sits inside a magnetic island. The islands move toward one another, compress the field between them, form a thin current sheet near an X-point, and reconnect at finite resistivity. This single case exercises all important version-1 terms:

- flux advection $[U, \psi]$;
- vorticity advection $[U, \omega]$;
- Lorentz forcing $[J, \psi]$;
- resistive diffusion $\eta\nabla_\perp^2\psi$;
- viscous diffusion $\nu\nabla_\perp^2\omega$;
- nonlinear transfer between magnetic and kinetic energy;
- current-sheet formation; and
- resistive change of magnetic topology.

The benchmark is adapted to the project's $[0, 2\pi)^2$ coordinates and sign conventions from the classical periodic coalescence problem used by Ng and collaborators [4, 5].

3.2 Exact initial fields

The fixed equilibrium amplitude is

$$\Psi_{\mathrm{eq}} = 1. \tag{3.1}$$

The version-1 initial condition is

$$\psi(x, y, 0) = \psi_0(x, y) = \Psi_{\mathrm{eq}} \sin x \sin y, \tag{3.2}$$

$$U(x, y, 0) = U_0(x, y) = A_0(\cos x - \cos y). \tag{3.3}$$

The parameter $A_0 \geq 0$ controls the coalescence-driving flow. In the full-field surrogate formulation, the network is supplied the entire numerical array $U_0(x, y)$ rather than only the scalar amplitude A_0. Thus the initial flow strength is already encoded point by point in the input field. The corresponding derived fields are

$$J_0 = -\nabla_\perp^2 \psi_0 = 2\Psi_{\mathrm{eq}} \sin x \sin y = 2\psi_0, \tag{3.4}$$

$$\omega_0 = \nabla_\perp^2 U_0 = -A_0(\cos x - \cos y) = -U_0, \tag{3.5}$$

$$\boldsymbol{B}_{\perp,0} = \Psi_{\mathrm{eq}}(\sin x \cos y, -\cos x \sin y), \tag{3.6}$$

$$\boldsymbol{v}_{\perp,0} = (-A_0 \sin y, -A_0 \sin x). \tag{3.7}$$

Verification checkpoint

Equations (3.4) and (3.5) are immediate sign tests. If an implementation produces $J_0 = -2\psi_0$ or $\omega_0 = +U_0$, its Fourier Laplacian or frozen field convention is wrong.

Important distinction : operator-generalization limit

Although the revised surrogate receives a complete two-dimensional U_0 field, version 1 still varies only the amplitude of the fixed shape $\cos x - \cos y$. The family $U_0^{(A_0)} = A_0(\cos x - \cos y)$ is therefore one-dimensional in initial-condition shape space. Success on this family does not establish generalization to arbitrary initial flow or magnetic fields. Broader operator learning would require additional, independently verified initial-field shapes.

3.3 Why the magnetic array is an equilibrium

Set $A_0 = 0$, so $U_0 = \omega_0 = 0$. Because

$$J_0 = 2\psi_0, \tag{3.8}$$

the Lorentz bracket vanishes:

$$[J_0, \psi_0] = [2\psi_0, \psi_0] = 0. \tag{3.9}$$

Thus the initial magnetic field produces no vorticity forcing in the ideal two-field model. When $\eta = 0$, it is a static exact equilibrium.

At finite η and $A_0 = 0$, the shape remains a single Fourier eigenmode:

$$\psi_{\text{exact}}(x, y, t) = \Psi_{\text{eq}} e^{-2\eta t} \sin x \sin y, \tag{3.10}$$

$$U_{\text{exact}}(x, y, t) = 0, \tag{3.11}$$

$$J_{\text{exact}}(x, y, t) = 2\psi_{\text{exact}}, \tag{3.12}$$

$$\omega_{\text{exact}}(x, y, t) = 0. \tag{3.13}$$

This exact resistively decaying solution is one of the strongest Module 2 verification cases.

3.4 O-points, X-points, and current channels

The magnetic critical points satisfy $\nabla\psi_0 = 0$. The four island centers in one fundamental domain are

Point	Coordinate	ψ_0/Ψ_{eq}	J_0/Ψ_{eq}
$O_{++}^{(1)}$	$(\pi/2, \pi/2)$	$+1$	$+2$
$O_{--}^{(1)}$	$(\pi/2, 3\pi/2)$	-1	-2
$O_{--}^{(2)}$	$(3\pi/2, \pi/2)$	-1	-2
$O_{++}^{(2)}$	$(3\pi/2, 3\pi/2)$	$+1$	$+2$

The points $(m\pi, n\pi)$ are X-points, with integers m, n. In particular,

$$X_c = (\pi, \pi) \tag{3.14}$$

lies between the two negative-current islands that the selected perturbation drives together.

3.5 How the perturbation drives coalescence

Evaluate Eq. (3.7) at the two negative-current O-points:

$$\boldsymbol{v}_{\perp,0}\left(\frac{\pi}{2}, \frac{3\pi}{2}\right) = (+A_0, -A_0), \tag{3.15}$$

$$\boldsymbol{v}_{\perp,0}\left(\frac{3\pi}{2}, \frac{\pi}{2}\right) = (-A_0, +A_0). \tag{3.16}$$

Both velocities point toward (π, π). Since both islands carry current in the same out-of-plane direction, the perturbation selects the attractive coalescence mode. The initial flow is not a random disturbance; its symmetry determines which pair merges inside the displayed fundamental cell.

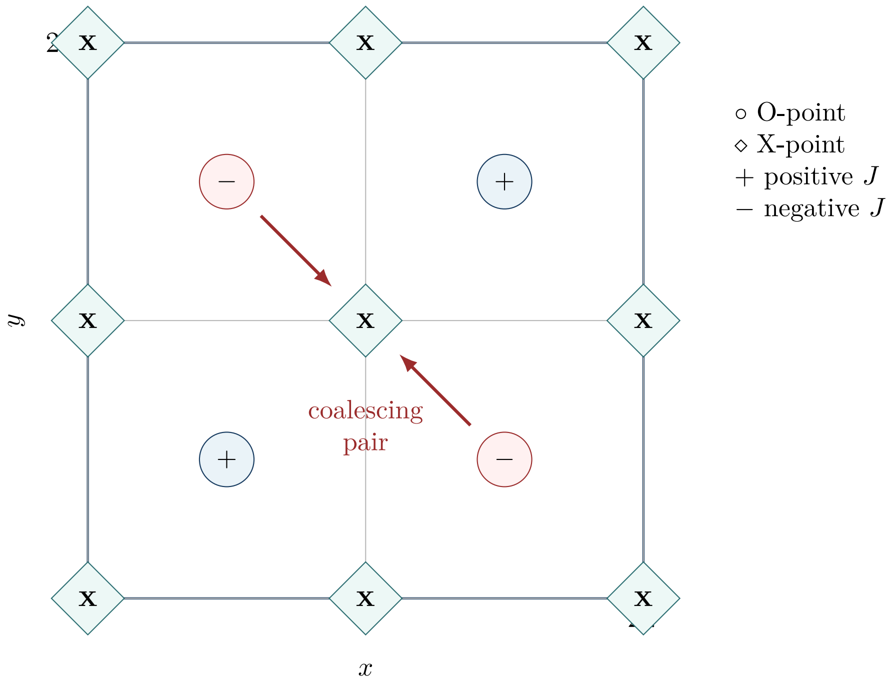

Figure 1: Topology of $\psi_0 = \sin x \sin y$. The stream-function perturbation drives the two negative-current islands toward the central X-point. Periodicity creates an equivalent pairing of the positive-current islands across the domain boundary. The drawing identifies critical points; it is not a computed trajectory.

The early qualitative sequence is

$$
\begin{aligned}
\text{two parallel current channels} &\longrightarrow \text{approach} \\
&\longrightarrow \text{magnetic compression at } X_c \\
&\longrightarrow \text{thin current sheet} \\
&\longrightarrow \text{finite } \eta \text{ changes } \psi \\
&\longrightarrow \text{reconnected flux and merged topology.}
\end{aligned}
$$

3.6 Symmetries as diagnostics

The analytic initial fields have reflection and exchange symmetries. Examples include

$$\psi_0(x, y) = \psi_0(y, x), \tag{3.17}$$

$$U_0(x, y) = -U_0(y, x), \tag{3.18}$$

$$\psi_0(2\pi - x, 2\pi - y) = \psi_0(x, y), \tag{3.19}$$

$$U_0(2\pi - x, 2\pi - y) = U_0(x, y). \tag{3.20}$$

The continuous equations preserve the compatible symmetry class. The solver does not need to impose these relations, but their defects reveal asymmetry from indexing, roundoff amplification, unequal grid treatment, or programming errors.

3.7 Baseline parameters and time horizon

The recommended baseline is

$$(\eta, \nu, A_0) = (10^{-3}, 10^{-3}, 0.1), \qquad \Psi_{\text{eq}} = 1. \tag{3.21}$$

The principal grid and output settings are

$$N_x = N_y = 128, \qquad \Delta t = 10^{-3}, \qquad \Delta t_{\text{save}} = 0.05. \tag{3.22}$$

The first pilot window is $0 \le t \le 5$. The accepted coalescence trajectory must extend through the first resolved current or reconnection-rate peak and include a post-peak interval. A nominal production horizon of $T = 8$ is therefore used for the baseline unless the convergence pilot demonstrates that a different common horizon is required.

Important distinction : endpoint

A final time is not physically adequate merely because the code reached it. If $|J|_{\max}$ and the reconnection rate are still rising at the last snapshot, the run has captured current-sheet formation but not the first coalescence event. Extend the horizon and version the configuration before generating the learning dataset.

4 Why a Fourier pseudo-spectral method

4.1 Periodic Fourier basis

A smooth doubly periodic field can be written

$$f(x, y, t) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \widehat{f}_{mn}(t) e^{i(k_{x,m}x + k_{y,n}y)}, \tag{4.1}$$

where

$$k_{x,m} = \frac{2\pi m}{L_x}, \qquad k_{y,n} = \frac{2\pi n}{L_y}. \tag{4.2}$$

For $L_x = L_y = 2\pi$, the wave numbers are integers:

$$k_{x,m} = m, \qquad k_{y,n} = n. \tag{4.3}$$

Spectral differentiation follows from

$$\partial_x e^{i(k_x x+k_y y)} = ik_x e^{i(k_x x+k_y y)}, \tag{4.4}$$
$$\partial_y e^{i(k_x x+k_y y)} = ik_y e^{i(k_x x+k_y y)}, \tag{4.5}$$
$$\nabla_\perp^2 e^{i(k_x x+k_y y)} = -(k_x^2+k_y^2) e^{i(k_x x+k_y y)}. \tag{4.6}$$

Thus

$$\widehat{f_x}_{mn} = ik_{x,m}\widehat{f}_{mn}, \tag{4.7}$$
$$\widehat{f_y}_{mn} = ik_{y,n}\widehat{f}_{mn}, \tag{4.8}$$
$$\widehat{\nabla_\perp^2 f}_{mn} = -k_{mn}^2\widehat{f}_{mn}, \tag{4.9}$$

where $k_{mn}^2 = k_{x,m}^2 + k_{y,n}^2$.

4.2 Discrete grid and transform

For even N_x, N_y, the collocation grid is

$$x_j = \frac{jL_x}{N_x}, \qquad j = 0, \ldots, N_x - 1, \tag{4.10}$$
$$y_\ell = \frac{\ell L_y}{N_y}, \qquad \ell = 0, \ldots, N_y - 1. \tag{4.11}$$

No duplicated endpoint is stored.

With the common unnormalized forward transform,

$$\widehat{f}_{mn} = \sum_{j=0}^{N_x-1}\sum_{\ell=0}^{N_y-1} f_{j\ell} e^{-2\pi i(mj/N_x + n\ell/N_y)}, \tag{4.12}$$
$$f_{j\ell} = \frac{1}{N_x N_y}\sum_{m=0}^{N_x-1}\sum_{n=0}^{N_y-1} \widehat{f}_{mn} e^{2\pi i(mj/N_x + n\ell/N_y)}. \tag{4.13}$$

This is the default normalization used by both NumPy's `fft2`/`ifft2` and MATLAB's `fft2`/`ifft2`. The stored wave-number ordering must follow the actual library ordering, including negative frequencies in the upper index range.

4.3 Why the method is called pseudo-spectral

Linear derivatives are evaluated in Fourier space. A nonlinear bracket, however, is evaluated by:

1. multiplying Fourier coefficients by ik_x or ik_y;
2. inverse transforming the derivatives to the physical grid;

3. multiplying physical-space arrays pointwise; and
4. transforming the product back to Fourier space.

The method is called *pseudo-spectral* because nonlinear products are evaluated at collocation points rather than by an explicit exact convolution sum over Fourier coefficients.

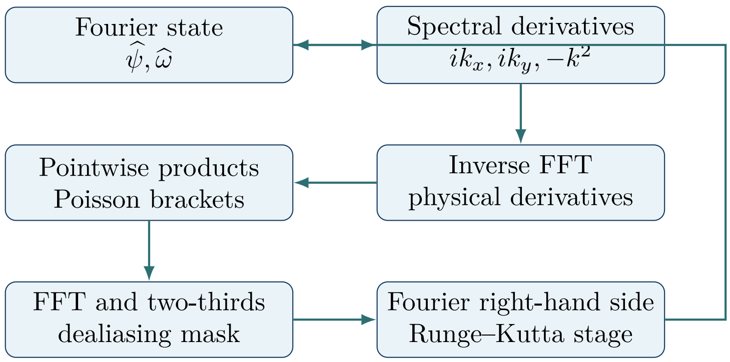

Figure 2: One right-hand-side evaluation in the Fourier pseudo-spectral solver. Derived fields and derivatives must be reconstructed from the current Runge–Kutta stage, not from the beginning of the time step.

4.4 Poisson inversion for the stream function

The equation $\nabla_\perp^2 U = \omega$ becomes

$$-k_{mn}^2 \widehat{U}_{mn} = \widehat{\omega}_{mn}. \tag{4.14}$$

For $k_{mn}^2 > 0$,

$$\boxed{\widehat{U}_{mn} = -\frac{\widehat{\omega}_{mn}}{k_{mn}^2}.} \tag{4.15}$$

At $(m,n) = (0,0)$, $k^2 = 0$, and division is undefined. The gauge and compatibility conditions set

$$\widehat{U}_{00} = 0, \qquad \widehat{\omega}_{00} = 0. \tag{4.16}$$

The current is simpler:

$$\boxed{\widehat{J}_{mn} = k_{mn}^2 \widehat{\psi}_{mn}.} \tag{4.17}$$

4.5 Hermitian symmetry and real fields

For a real physical field,

$$\widehat{f}(-\boldsymbol{k}) = \widehat{f}(\boldsymbol{k})^*, \tag{4.18}$$

where the star denotes complex conjugation. Roundoff produces tiny imaginary parts after inverse transformation. The implementation may discard those parts only after checking that

$$\frac{\|\mathrm{Im}\, f\|_2}{\max(\|\mathrm{Re}\, f\|_2\,, \epsilon_{\mathrm{mach}})} \tag{4.19}$$

remains near roundoff. A large imaginary fraction signals broken mode indexing or Hermitian symmetry.

5 Nonlinear brackets and aliasing

5.1 Fourier-space semi-discrete equations

Define $\widehat{\mathcal{B}(f,g)}$ as the dealiased transform of $[f,g]$. Then

$$\frac{\mathrm{d}\widehat{\psi}}{\mathrm{d}t} = -\widehat{\mathcal{B}(U,\psi)} - \eta k^2 \widehat{\psi}, \tag{5.1}$$

$$\frac{\mathrm{d}\widehat{\omega}}{\mathrm{d}t} = -\widehat{\mathcal{B}(U,\omega)} + \widehat{\mathcal{B}(J,\psi)} - \nu k^2 \widehat{\omega}. \tag{5.2}$$

At each evaluation:

$$\widehat{U} = -\widehat{\omega}/k^2 \quad (k \neq 0), \tag{5.3}$$

$$\widehat{J} = k^2 \widehat{\psi}. \tag{5.4}$$

5.2 Physical-space bracket evaluation

For two fields f and g,

$$[f,g] = f_x g_y - f_y g_x. \tag{5.5}$$

A direct evaluation uses four inverse transforms:

$$f_x = \mathcal{F}^{-1}(ik_x\widehat{f}), \qquad f_y = \mathcal{F}^{-1}(ik_y\widehat{f}), \tag{5.6}$$

$$g_x = \mathcal{F}^{-1}(ik_x\widehat{g}), \qquad g_y = \mathcal{F}^{-1}(ik_y\widehat{g}), \tag{5.7}$$

followed by

$$\widehat{\mathcal{B}(f,g)} = \mathcal{P}_{2/3}\mathcal{F}(f_x g_y - f_y g_x). \tag{5.8}$$

Here $\mathcal{P}_{2/3}$ is the dealiased spectral projection.

5.3 What aliasing means

Suppose two resolved one-dimensional modes have wave numbers p and q. Their product contains wave number $p+q$. If $|p+q|$ exceeds the largest representable wave number, the sampled grid cannot distinguish it from a lower wave number separated by an integer multiple of N. The unresolved contribution is then folded, or *aliased*, into a resolved mode.

Aliasing is especially dangerous in nonlinear MHD because it can:

- move energy into incorrect modes;
- violate discrete versions of bracket identities;
- contaminate current and vorticity more strongly than ψ and U;
- mimic numerical instability or artificial dissipation; and
- create false small islands near an underresolved current sheet.

5.4 Two-thirds truncation

For quadratic nonlinearities, the version-1 solver retains only modes satisfying

$$|m| \le \left\lfloor \frac{N_x}{3} \right\rfloor, \qquad |n| \le \left\lfloor \frac{N_y}{3} \right\rfloor, \tag{5.9}$$

with a consistent decision at the boundary index in both languages. All other coefficients are set to zero after each nonlinear transform and after every completed stage update.

Frozen Module 2 convention

The project uses the rectangular two-thirds mask in Eq. (5.9). A circular mask, exponential filter, phase-shift method, or three-halves padding method is a different numerical contract and must be versioned rather than silently substituted.

5.5 Why filtering only the saved output is insufficient

Aliased terms affect the time derivative before a snapshot is written. Removing high modes only at output time cannot undo their dynamical effect. Dealiasing is part of every nonlinear right-hand-side evaluation.

5.6 Bracket identity checks

For smooth periodic fields,

$$[f, f] = 0, \tag{5.10}$$

$$[f, g] = -[g, f], \tag{5.11}$$

$$\int_\Omega [f, g] \, \mathrm{d}A = 0, \tag{5.12}$$

$$\int_\Omega f [g, h] \, \mathrm{d}A = \int_\Omega g [h, f] \, \mathrm{d}A. \tag{5.13}$$

The discrete dealiased operator should satisfy these identities to roundoff for resolved test modes. Their failure is more diagnostic than merely comparing one bracket array with a finite-difference estimate.

6 Classical fourth-order Runge–Kutta integration

6.1 Method-of-lines system

After spatial discretization, collect the Fourier coefficients into

$$\boldsymbol{q}(t) = \begin{bmatrix} \widehat{\psi}(t) \\ \widehat{\omega}(t) \end{bmatrix}, \qquad \frac{\mathrm{d}\boldsymbol{q}}{\mathrm{d}t} = \boldsymbol{\mathcal{L}}(\boldsymbol{q}; \eta, \nu). \tag{6.1}$$

The operator $\boldsymbol{\mathcal{L}}$ includes Poisson inversion, current reconstruction, dealiased brackets, and diffusion.

6.2 RK4 stages

For step size Δt, classical fourth-order Runge–Kutta (RK4) is

$$\boldsymbol{k}_1 = \boldsymbol{\mathcal{L}}(\boldsymbol{q}^n), \tag{6.2}$$

$$\boldsymbol{k}_2 = \boldsymbol{\mathcal{L}}\left(\boldsymbol{q}^n + \frac{\Delta t}{2}\boldsymbol{k}_1\right), \tag{6.3}$$

$$\boldsymbol{k}_3 = \boldsymbol{\mathcal{L}}\left(\boldsymbol{q}^n + \frac{\Delta t}{2}\boldsymbol{k}_2\right), \tag{6.4}$$

$$\boldsymbol{k}_4 = \boldsymbol{\mathcal{L}}\left(\boldsymbol{q}^n + \Delta t\, \boldsymbol{k}_3\right), \tag{6.5}$$

$$\boldsymbol{q}^{n+1} = \boldsymbol{q}^n + \frac{\Delta t}{6}\left(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4\right). \tag{6.6}$$

For sufficiently smooth solutions and fixed spatial discretization, the global time-discretization error is $O(\Delta t^4)$.

6.3 Derived fields must be stage-consistent

Every evaluation of $\boldsymbol{\mathcal{L}}$ receives a different stage approximation. Therefore each stage must:

1. apply the spectral mask to the stage state;
2. set the vorticity zero mode to zero;
3. reconstruct $\widehat{U}$ from the stage $\widehat{\omega}$;
4. reconstruct $\widehat{J}$ from the stage $\widehat{\psi}$;
5. compute stage derivatives and brackets;
6. apply the dealiasing mask to nonlinear transforms; and
7. return the Fourier right-hand side.

Reusing U or J from the beginning of the step reduces accuracy and changes the algorithm.

6.4 Fixed time step and stability monitors

Version 1 uses a fixed Δt within each convergence run. This simplifies parity and observed-order analysis. The chosen step must remain below conservative advective/Alfvénic and diffusive limits. Useful monitors are

$$\Delta t_{\text{adv}} = C_{\text{adv}} \min\left[\frac{\Delta x}{\max(|v_x| + |B_x|)}, \frac{\Delta y}{\max(|v_y| + |B_y|)}\right], \tag{6.7}$$

$$\Delta t_\eta = \frac{C_{\text{diff}}}{\eta k_{\max}^2}, \tag{6.8}$$

$$\Delta t_\nu = \frac{C_{\text{diff}}}{\nu k_{\max}^2}, \tag{6.9}$$

where $k_{\max}$ is the largest retained dealiased wave-number magnitude. The constants C_{adv} and C_{diff} are safety factors, not physical parameters. The code records the minimum ratios $\Delta t/\Delta t_{\text{adv}}$, $\Delta t/\Delta t_\eta$, and $\Delta t/\Delta t_\nu$ across the run.

Important distinction

Satisfying a stability estimate does not prove accuracy. A stable time step may still shift the time of peak reconnection or damp a thin current sheet. Temporal refinement is required even when no instability occurs.

6.5 Why RK4 is the version-1 choice

RK4 is explicit, familiar, independently implementable, and fourth-order accurate for smooth solutions. It avoids hiding differences behind library-specific adaptive controllers. The baseline coefficients and grid permit a conservative explicit step. If later sweeps require substantially smaller diffusivities or much higher resolution, an integrating-factor, exponential, or implicit–explicit method may be more efficient; adopting one would require a new time-integration verification and a recorded method version.

7 Complete solver algorithm

7.1 Initialization

1. Read and validate the case configuration.
2. Create half-open x and y grids.
3. Create FFT-ordered k_x, k_y, and k^2 arrays.
4. Create the rectangular two-thirds mask.
5. Evaluate Eqs. (3.2)–(3.3).
6. Transform ψ_0 and U_0 to Fourier space.
7. Set $\widehat{\omega}_0 = -k^2\widehat{U}_0$.
8. Project the initial state with the dealiasing mask.
9. Enforce $\widehat{\omega}_{00} = 0$ and the recorded $\widehat{\psi}_{00}$.
10. Reconstruct U_0, J_0, ω_0 and run the analytic initial-condition checks.

7.2 One time step

1. Evaluate four stage right-hand sides using Eq. (6.6).
2. Apply the spectral mask to the updated $\widehat{\psi}$ and $\widehat{\omega}$.
3. Restore the prescribed mean-flux coefficient and zero vorticity mode.
4. Check that all coefficients are finite.
5. Evaluate time-step safety monitors.
6. If this is a save time, reconstruct all fields and diagnostics from the same state.

7.3 One save event

At a saved time t_n, write:

- ψ, U, J, ω on the common physical grid;
- magnetic and kinetic energies;
- Ohmic and viscous dissipation rates;
- peak absolute current and vorticity;
- O-point and X-point locations and flux values;
- reconnected-flux and reconnection-rate estimates;
- periodicity, divergence, mean, symmetry, and definition defects;
- high-wave-number spectral-tail measures; and
- accumulated wall-clock and transform counts.

7.4 Termination

A run terminates successfully only if it reaches the requested final time with finite fields and no failed acceptance monitor. A process exit without an exception is not automatically a scientifically successful case. The manifest records separate booleans for execution, verification, convergence, and dataset acceptance.

8 Physical and numerical diagnostics

8.1 Field amplitudes

Store

$$\|f\|_2 = \left(\int_\Omega f^2 \,\mathrm{d}A\right)^{1/2}, \tag{8.1}$$

$$\|f\|_\infty = \max_{(x,y)\in\Omega} |f(x,y)|, \tag{8.2}$$

for $f \in \{\psi, U, J, \omega\}$. The derived fields are particularly sensitive to small-scale error:

$$J = -\nabla_\perp^2 \psi, \qquad \omega = \nabla_\perp^2 U. \tag{8.3}$$

A small error in ψ does not guarantee a small error in J.

8.2 Energy and integrated dissipation

Define

$$\mathcal{D}_\eta(t) = \eta \int_\Omega J^2 \, \mathrm{d}A, \tag{8.4}$$

$$\mathcal{D}_\nu(t) = \nu \int_\Omega \omega^2 \, \mathrm{d}A. \tag{8.5}$$

The integrated energy-balance defect is

$$\delta_E(t) = \frac{\left| E(t) - E(0) + \int_0^t [\mathcal{D}_\eta(s) + \mathcal{D}_\nu(s)] \, \mathrm{d}s \right|}{\max(E(0), \epsilon_{\mathrm{mach}})}. \tag{8.6}$$

The time integral should use a quadrature consistent with the available diagnostic samples. For stringent verification, evaluate dissipation at RK stages or save it at every time step rather than only at sparse output times.

8.3 Mean-square flux and cross helicity

Module 1 derived

$$\mathcal{A}(t) = \frac{1}{2} \int_\Omega \psi^2 \, \mathrm{d}A, \tag{8.7}$$

$$\frac{\mathrm{d}\mathcal{A}}{\mathrm{d}t} = -2\eta E_B, \tag{8.8}$$

$$H_C(t) = \int_\Omega \psi\omega \, \mathrm{d}A, \tag{8.9}$$

$$\frac{\mathrm{d}H_C}{\mathrm{d}t} = -(\eta + \nu) \int_\Omega J\omega \, \mathrm{d}A. \tag{8.10}$$

These independent integral identities strengthen the verification suite because they weight the fields differently from total energy.

8.4 Divergence defects

Although the scalar-potential representation makes divergence vanish analytically, compute

$$\delta_{\nabla\cdot B} = \|\partial_x B_x + \partial_y B_y\|_\infty \,, \tag{8.11}$$

$$\delta_{\nabla\cdot v} = \|\partial_x v_x + \partial_y v_y\|_\infty \,. \tag{8.12}$$

These should be near roundoff when both components use the same spectral derivatives.

8.5 Definition defects

When fields are reconstructed for storage, evaluate

$$\delta_J = \frac{\|J + \nabla_\perp^2 \psi\|_2}{\max(\|J\|_2\,, \epsilon_{\mathrm{mach}})}, \tag{8.13}$$

$$\delta_\omega = \frac{\|\omega - \nabla_\perp^2 U\|_2}{\max(\|\omega\|_2\,, \epsilon_{\mathrm{mach}})}. \tag{8.14}$$

If J and ω are generated from the same Fourier coefficients, these are roundoff checks. They become important after output conversion, interpolation, compression, or neural prediction.

8.6 Periodicity defects

Because the endpoint is not duplicated, periodicity is best checked spectrally or by evaluating the represented field at matched faces. For a diagnostic evaluator,

$$\delta_{x,f}^{(r)} = \max_y |\partial_x^r f(0, y) - \partial_x^r f(2\pi, y)|\,, \tag{8.15}$$

$$\delta_{y,f}^{(r)} = \max_x \left|\partial_y^r f(x, 0) - \partial_y^r f(x, 2\pi)\right|. \tag{8.16}$$

Use derivative orders required by the eventual formulation, not values alone.

8.7 Symmetry defects

Examples are

$$\delta_{\psi,\mathrm{exch}} = \frac{\|\psi(x, y) - \psi(y, x)\|_2}{\max(\|\psi\|_2\,, \epsilon_{\mathrm{mach}})}, \tag{8.17}$$

$$\delta_{U,\mathrm{exch}} = \frac{\|U(x, y) + U(y, x)\|_2}{\max(\|U\|_2\,, \epsilon_{\mathrm{mach}})}. \tag{8.18}$$

The parity of J follows ψ, while the parity of ω follows U.

8.8 Current-sheet diagnostics

At each saved time, record:

$$J_{\max}(t) = \|J(\cdot, \cdot, t)\|_\infty\,, \tag{8.19}$$

$$(x_J(t), y_J(t)) = \arg\max_{x,y} |J(x, y, t)|. \tag{8.20}$$

Near the central sheet, also estimate:

- sheet orientation from the local Hessian or a prescribed symmetry line;
- full width at half maximum of $|J|$ across the sheet;
- sheet length along the high-current ridge;
- number of grid intervals across the half-maximum width; and
- maximum gradients of J and ω.

An absolute peak is grid-sensitive. It must be paired with sheet width, spectra, and resolution refinement.

8.9 Locating O-points and X-points

A magnetic critical point satisfies

$$\nabla\psi = 0. \tag{8.21}$$

Let

$$H_\psi = \begin{bmatrix} \psi_{xx} & \psi_{xy} \\ \psi_{xy} & \psi_{yy} \end{bmatrix} \tag{8.22}$$

be the Hessian. Its determinant classifies a nondegenerate critical point:

$$\det H_\psi > 0: \quad \text{O-point,} \tag{8.23}$$

$$\det H_\psi < 0: \quad \text{X-point.} \tag{8.24}$$

The algorithm should:

1. find grid candidates where $|\nabla\psi|$ is locally small;
2. refine each location by Newton iteration using spectral gradients and Hessians;
3. classify the point;
4. associate it with the nearest previous-time point; and
5. record loss, creation, or ambiguous association explicitly.

Fixed-coordinate flux values are useful symmetry checks, but moving critical points are required for robust topology diagnostics.

8.10 Reconnected flux

For one of the two merging negative-current islands, let $\psi_O(t)$ be the flux at its tracked O-point and $\psi_X(t)$ the flux at the separating central X-point. Define the remaining O-to-X flux contrast

$$\Delta\psi_{OX}(t) = \psi_X(t) - \psi_O(t). \tag{8.25}$$

Initially, $\Delta\psi_{OX}(0) = \Psi_{\mathrm{eq}}$ for the exact points. Define the signed reconnected flux by

$$\Phi_{\mathrm{rec}}(t) = \Delta\psi_{OX}(0) - \Delta\psi_{OX}(t), \tag{8.26}$$

and the signed reconnection rate by

$$\mathcal{R}_{\mathrm{rec}}(t) = \frac{\mathrm{d}\Phi_{\mathrm{rec}}}{\mathrm{d}t}. \tag{8.27}$$

At a smoothly moving critical point $\boldsymbol{x}_c(t)$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\psi(\boldsymbol{x}_c(t), t) = \partial_t\psi + \dot{\boldsymbol{x}}_c \cdot \nabla\psi = \partial_t\psi, \tag{8.28}$$

because $\nabla\psi = 0$. Therefore,

$$\mathcal{R}_{\mathrm{rec}} = -\left[\partial_t\psi_X - \partial_t\psi_O\right]. \tag{8.29}$$

The PDE right-hand side supplies a less noisy derivative than finite differencing sparse saved values. Report both the signed value and its magnitude; do not hide sign changes during sloshing.

Important distinction : definition

The phrase *reconnected flux* is not unique without specifying the O-point, X-point, sign, reference value, and tracking rule. Every plot and stored array must use Eqs. (8.25)–(8.27) or declare a versioned alternative.

8.11 Spectral diagnostics

Define a shell-averaged magnetic spectrum, for example,

$$E_B(\kappa) = \frac{1}{2} \sum_{\kappa-1/2<|\boldsymbol{k}|\leq\kappa+1/2} k^2 |\widehat{\psi}_{\boldsymbol{k}}|^2 \times C_{\mathrm{FFT}}, \tag{8.30}$$

where C_{FFT} is the normalization consistent with Parseval's identity. Define $E_K(\kappa)$ similarly using $\widehat{U}$.

Useful resolution measures include:

$$\rho_{B,\mathrm{tail}} = \frac{\sum_{\kappa\geq 0.8\kappa_{\mathrm{cut}}} E_B(\kappa)}{\sum_\kappa E_B(\kappa)}, \tag{8.31}$$

$$\rho_{K,\text{tail}} = \frac{\sum_{\kappa \geq 0.8\kappa_{\text{cut}}} E_K(\kappa)}{\sum_{\kappa} E_K(\kappa)}. \tag{8.32}$$

A pile-up near the dealiased cutoff indicates underresolution even if the run remains finite.

9 Code-verification hierarchy

9.1 Test 1: Fourier derivatives

For

$$f(x, y) = \sin(mx)\cos(ny), \tag{9.1}$$

compare numerical derivatives with

$$f_x = m\cos(mx)\cos(ny), \tag{9.2}$$

$$f_y = -n\sin(mx)\sin(ny), \tag{9.3}$$

$$\nabla_\perp^2 f = -(m^2 + n^2)f. \tag{9.4}$$

Test positive, negative, mixed, zero, and near-cutoff retained modes. Errors for exactly represented low modes should be near roundoff.

9.2 Test 2: Poisson inversion

Choose a zero-mean analytic U, compute $\omega = \nabla_\perp^2 U$, invert Eq. (4.15), and compare with U. Add a constant to the analytic input to confirm that the solver returns the zero-mean gauge rather than attempting to recover an unobservable constant.

9.3 Test 3: analytic initial fields

For Eqs. (3.2)–(3.3), verify

$$J_0 = 2\psi_0, \qquad \omega_0 = -U_0, \tag{9.5}$$

and verify the analytic expressions for $\boldsymbol{B}_{\perp,0}$ and $\boldsymbol{v}_{\perp,0}$.

9.4 Test 4: bracket identities

Use resolved trigonometric fields to test antisymmetry, self-brackets, zero mean, and cyclic integration. Include at least one case in which the analytic bracket is nonzero.

9.5 Test 5: exact resistive decay

Set $A_0 = 0$ and compare with Eq. (3.13). This tests Fourier initialization, the magnetic Laplacian sign, resistive decay, RK4 temporal order, preservation of $U = \omega = 0$, and the energy and mean-square-flux identities.

9.6 Test 6: manufactured nonlinear solution

Choose smooth periodic manufactured fields, for example,

$$\psi^\star(x,y,t) = e^{-t}\left[\sin x \sin y + 0.2\cos(2x-y)\right], \tag{9.6}$$

$$U^\star(x,y,t) = 0.3e^{-2t}\left[\cos x - \cos y + 0.25\sin(x+2y)\right]. \tag{9.7}$$

Derive $J^\star = -\nabla_\perp^2\psi^\star$ and $\omega^\star = \nabla_\perp^2 U^\star$, and add test-only sources

$$S_\psi = \partial_t\psi^\star + [U^\star, \psi^\star] - \eta\nabla_\perp^2\psi^\star, \tag{9.8}$$

$$S_\omega = \partial_t\omega^\star + [U^\star, \omega^\star] - [J^\star, \psi^\star] - \nu\nabla_\perp^2\omega^\star. \tag{9.9}$$

With these sources, $(\psi^\star, U^\star)$ is exact. This test exercises every nonlinear term with known truth. The production solver remains unforced; source injection is a verification-only interface.

9.7 Test 7: ideal invariants

Set $\eta = \nu = 0$ for a short smooth run and verify that total energy, mean-square flux, and cross helicity converge toward constant histories as Δt and grid spacing are refined. Do not extend an explicit ideal run into an unresolved current-sheet regime and interpret failure as an operator bug.

9.8 Test 8: dissipative balances

For finite η, ν, evaluate Eq. (8.6) and the corresponding integrated defects for $\mathcal{A}$ and H_C. Confirm nonincreasing total energy. A small balance defect together with an increasing total energy is impossible under the frozen model and signals a diagnostic-sign error.

9.9 Test 9: symmetry, means, and divergence

Track the exact zero modes, exchange symmetries, and divergence defects. These tests are cheap and should run for every case, not only in a separate verification program.

9.10 Test 10: independent-language parity

Python and MATLAB must start from the same configuration and save the same requested times. Compare initial fields and Fourier coefficients, one right-hand-side evaluation, one complete RK4 step, a short nonlinear trajectory, diagnostic histories, and the baseline trajectory with identical array ordering. Roundoff-level agreement is expected for initialization and operators. Long nonlinear trajectories may differ slightly because of transform and floating-point ordering; differences must decrease under refinement and remain below the physics-metric tolerances.

10 Temporal and spatial convergence

10.1 Why refinement must be separated

The numerical error contains both spatial and temporal contributions:

$$e_{\text{total}} \approx e_{\text{space}}(N_x, N_y) + e_{\text{time}}(\Delta t) + e_{\text{roundoff}}. \tag{10.1}$$

If space and time are refined simultaneously, an apparent plateau or slope cannot be attributed cleanly. Module 2 therefore conducts separate studies.

10.2 Temporal refinement

Fix a spatial grid fine enough that spatial error is smaller than the anticipated time error. Use

$$\Delta t \in \{4, 2, 1, 0.5\} \times 10^{-3} \tag{10.2}$$

for the baseline pilot, subject to stability. Compare solutions at common physical times.

For a diagnostic Q, define a three-level self-convergence estimate

$$p_{\text{obs}} = \log_2 \left(\frac{|Q_{\Delta t} - Q_{\Delta t/2}|}{|Q_{\Delta t/2} - Q_{\Delta t/4}|} \right). \tag{10.3}$$

For a field, replace absolute values by a discrete norm. RK4 should approach $p_{\text{obs}} = 4$ in the asymptotic regime.

10.3 Spatial refinement

Fix a sufficiently small time step and run

$$N_x = N_y \in \{32, 64, 128, 256\}. \tag{10.4}$$

Compare coarse and fine solutions on exactly common grid points, or use a documented Fourier restriction of the fine solution. Do not compare interpolated plots.

For smooth fields, Fourier error may decrease faster than any fixed algebraic power until the current sheet becomes the limiting structure. Once a sheet is only marginally resolved, convergence may slow sharply. Therefore the spatial study reports error values and spectra rather than forcing a single polynomial order onto the full nonlinear trajectory.

10.4 Field self-convergence

Let $\mathcal{R}_{2N\to N}$ denote exact spectral restriction from a $2N$ grid to the modes represented on an N grid. Define

$$e_N^{(f)}(t) = \frac{\|f_N(t) - \mathcal{R}_{2N\to N} f_{2N}(t)\|_2}{\max(\|\mathcal{R}_{2N\to N} f_{2N}(t)\|_2\,, \epsilon_{\mathrm{mach}})}. \tag{10.5}$$

Compute this for $f \in \{\psi, U, J, \omega\}$ and report both the maximum over time and values near the reconnection peak.

10.5 Diagnostic convergence

At minimum, refine:

- $E_B(t)$, $E_K(t)$, and integrated dissipation;
- $J_{\max}(t)$ and its time and location;
- $\Phi_{\mathrm{rec}}(t)$;
- peak $|\mathcal{R}_{\mathrm{rec}}|$ and its time;
- current-sheet width at the peak; and
- topology-event times.

Peak values and event times are usually less convergent than global energy. A solver is not accepted because energy alone converged.

10.6 Resolution criteria

The default acceptance targets for a baseline case are:

1. the two finest spatial grids differ by less than 0.5% in ψ and U relative L_2 error at all saved times;

2. the two finest grids differ by less than 2% in peak current, peak reconnection rate, and their occurrence times;
3. at least eight grid intervals span the measured current-sheet full width at half maximum on the accepted grid;
4. magnetic and kinetic spectral tails decay without a pile-up at the dealiased cutoff;
5. the two finest time steps show the RK4 asymptotic trend on the exact-decay or manufactured test;
6. the baseline nonlinear diagnostics change by less than the stated tolerances under time-step halving; and
7. integrated energy-balance, means, definitions, divergence, and symmetry defects remain within their recorded numerical tolerances.

These are project acceptance criteria, not universal plasma-physics constants. If a case fails, increase resolution, shorten the parameter range, or reject the case. Do not loosen criteria after seeing an inconvenient result without recording and justifying the revision.

10.7 Error plateaus

Refinement eventually reaches a plateau caused by another error source. Time error may give way to spatial error, spatial error may give way to time error, and both may eventually give way to roundoff. Field L_2 error can converge while $J_{\max}$ remains underresolved, and balance laws can appear converged while topology-event times drift. The plateau must be diagnosed, not merely described as convergence.

11 Case generation, parameter design, and initial-field diversity

11.1 One verified baseline before a sweep

The order of work is:

1. pass numerical-operator and exact-solution tests;
2. converge one baseline nonlinear trajectory;
3. freeze its accepted numerical settings;
4. run a small parameter pilot;
5. determine which cases require higher resolution or a longer horizon;

6. generate the full sweep; and
7. freeze whole-case train, validation, and test assignments in Module 3 before extracting time-conditioned operator pairs.

Generating dozens of unverified runs first only multiplies uncertainty.

11.2 Initial parameter sweep

A compact first family is

$$\eta \in \{5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}, \tag{11.1}$$

$$\nu \in \{5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}, \tag{11.2}$$

$$A_0 \in \{0.05, 0.10, 0.20\}. \tag{11.3}$$

This produces 27 candidate physical trajectories spanning magnetic Prandtl numbers

$$\mathrm{Pm} = \frac{\nu}{\eta} \tag{11.4}$$

from 0.25 to 4. The sweep is a candidate design, not an automatic acceptance list. Lower diffusivity and larger perturbation may require finer grids and smaller time steps.

For operator learning, these 27 trajectories are the primary count of physical-case diversity. Each accepted trajectory can later supply several targets at different saved times, but those targets share the same initial state and dynamical history and must not be counted as independent physical cases.

11.3 Initial-field diversity under the full-field map

Equation (1.4) allows an input to be an entire spatial field rather than a small list of scalar parameters. That additional representational capacity does not create new physical diversity automatically. In version 1,

$$\psi_0 = \sin x \sin y, \qquad U_0 = A_0(\cos x - \cos y), \tag{11.5}$$

so ψ_0 has one fixed shape and U_0 has one fixed shape with three trial amplitudes. The first operator proof of concept therefore tests whether a field-to-field model can learn this controlled family, not whether it can evolve arbitrary reduced-MHD initial conditions.

A later, broader solution-operator study may introduce additional verified initial-field families, for example by varying mode content or topology. Such changes alter the physical case definition and must be versioned, converged, and validated numerically before those trajectories enter the learning dataset. They should not be introduced merely as neural-network augmentation.

11.4 Dimensionless interpretation

Under the Module 1 Alfvén normalization,

$$\eta = \frac{1}{S}, \qquad \nu = \frac{1}{\mathrm{Re}}, \qquad \mathrm{Pm} = \frac{\nu}{\eta}. \tag{11.6}$$

Here S and Re are based on the frozen reference scales, not on a sheet thickness measured later. The local sheet Lundquist number is a diagnostic derived from local length and field scales and should not silently replace the case parameter.

11.5 Randomness

The analytic version-1 initial condition is deterministic. The field arrays should not depend on the random seed. A seed is still stored for:

- future optional perturbation noise;
- case-split generation;
- operator-pair, batch, or time-target sampling during neural training; and
- reproducibility of any stochastic diagnostic procedure.

Noise is not added to the baseline coalescence field. At high Lundquist number, small noise can seed secondary islands and change reconnection dynamics [5]; it is therefore a physical-case parameter, not an innocuous numerical detail.

11.6 Common output times and operator targets

Cases should save a common time grid when possible:

$$t_n = n\Delta t_{\mathrm{save}}. \tag{11.7}$$

The integrator must land exactly on requested output times, either because $\Delta t_{\mathrm{save}}/\Delta t$ is an integer or through a documented final substep. Interpolating a saved field in time changes the reference data and must be recorded.

For an accepted trajectory, any stored time t_n can later define a time-conditioned full-field pair

$$(\psi_0, U_0, \eta, \nu, t_n) \longmapsto (\psi_n, U_n). \tag{11.8}$$

Module 2 stores the complete trajectory rather than duplicating these pairs on disk. Module 3 decides which times become learning examples and how the scalar context is packed for the neural operator.

11.7 Event coverage across parameters

Changing η, ν, or A_0 changes the time of current-sheet formation and peak reconnection. A fixed T may capture a full event for one case and only the linear phase for another. Store both physical time and event markers:

- current-sheet-onset time;
- time of first $J_{\max}$ peak;
- time of first reconnection-rate peak;
- first topology merger time; and
- whether a post-peak interval is present.

The primary operator may use physical time t as its conditioning variable, while event markers remain essential for interpreting unequal dynamics and constructing balanced evaluation sets.

12 Data, provenance, and full-field operator handoff

12.1 Minimum HDF5 contents

The preferred authoritative field container is HDF5. A case must contain:

$$\begin{aligned} \text{coordinates: } & x[N_x],\ y[N_y],\ t[N_t], && (12.1)\\ \text{fields: } & \psi[N_t,N_x,N_y],\ U[N_t,N_x,N_y], && (12.2)\\ & J[N_t,N_x,N_y],\ \omega[N_t,N_x,N_y], && (12.3)\\ \text{parameters: } & \eta,\nu,A_0,\Psi_{\text{eq}}, && (12.4)\\ \text{diagnostics: } & \text{time-aligned scalar and event histories.} && (12.5) \end{aligned}$$

The actual on-disk axis order must be explicit and identical in metadata. MATLAB and NumPy use different natural memory and indexing conventions; reshaping without an axis contract can transpose the physics silently.

12.2 Numerical fields are not rendered images

For the full-field surrogate, the learning object is the numerical scalar array itself. At a fixed saved time,

$$\psi_n = \{\psi(x_i,y_j,t_n)\}_{i,j}, \qquad U_n = \{U(x_i,y_j,t_n)\}_{i,j}. \tag{12.6}$$

These arrays preserve sign, amplitude, coordinate registration, floating-point precision, and exact correspondence between variables.

A contour plot, magnetic-field-line rendering, velocity-vector plot, streamline image, PNG, JPEG, or screenshot is only a visualization of those values. Rendering introduces choices such as color map, pixel resolution, antialiasing, line thickness, contour levels, axes, and labels and is generally many-to-one. Such images are useful for human inspection but are excluded from the surrogate training and validation data path.

Important distinction : no image surrogate data

The revised project is a full-field numerical-array learning problem, not a computer-vision problem. Every operator input and target must be traceable directly to accepted numerical field arrays. A rendered image may accompany a case for visualization but may not replace the underlying $N_x \times N_y$ scalar data.

12.3 Operator-ready view of one accepted trajectory

One accepted HDF5 trajectory can be viewed downstream as a collection of field-to-field examples. For each selected saved time t_n, define

$$X_n = (\psi[0,:,:], U[0,:,:], \eta, \nu, t_n), \tag{12.7}$$

$$Y_n = (\psi[n,:,:], U[n,:,:]). \tag{12.8}$$

Equivalently, $X_n \mapsto Y_n$ is the discrete-grid realization of Eq. (1.4). The initial fields and targets remain separate numerical channels; Module 2 does not concatenate or rescale them irreversibly.

The scalar context variables η, ν, and t_n are stored separately. A later architecture may feed them through a separate branch, broadcast them as constant spatial channels, or encode them in another reversible representation. Likewise, x and y remain explicit coordinate arrays so that Module 3 can add coordinate channels, periodic features, or spectral-grid information without modifying the reference data.

For the version-1 analytic family, A_0 is already encoded in U_0. It therefore need not be a duplicate neural input in the primary operator formulation. It remains mandatory case metadata because it defines how the simulation was generated, supports stratified reporting, and provides a direct consistency check between the stored parameter and initial field.

12.4 Whole trajectories remain the independence unit

A single trajectory with N_t stored times can generate up to N_t pairs of the form in Eq. (11.8), but those pairs share one initial condition, one parameter case, and one numerical evolution. They are strongly dependent. Module 2 therefore publishes complete trajectories and case identifiers, not a shuffled table of field pairs.

Module 3 must assign whole accepted trajectories to train, validation, and test partitions before selecting target times. This prevents a model from seeing early or neighboring snapshots from a trajectory during training and then being evaluated on later snapshots from that same nominally "held-out" case while claiming unseen-case generalization.

12.5 Required metadata

Store:

- case identifier and schema version;
- creation timestamp;
- source revision or content hash;
- equations-and-conventions version;
- initial-condition name and formula version;
- full-field operator-interface version;
- L_x, L_y, N_x, N_y and explicit axis order;
- FFT ordering and normalization;
- dealiasing method and retained-mode rule;
- time integrator, Δt, final time, and save interval;
- physical normalization and parameter values;
- floating-point precision;
- language and dependency versions;
- execution platform;
- verification and convergence report references;
- all acceptance flags; and
- warning or rejection reasons.

12.6 Field precision and compression

Generate and verify in 64-bit floating point. Lossless compression may be used after byte-for-byte readback tests. Do not convert accepted reference fields to single precision without quantifying the resulting errors in ψ, U, J, ω, PDE residuals, and diagnostics. Do not replace numerical arrays with rendered image formats for storage reduction.

12.7 Atomic output

Write a case first to a temporary filename, validate its contents, flush and close it, and then publish the completed file atomically. An interrupted run must not leave an incomplete file with a valid-looking production name.

12.8 Rejected cases

A rejected case may be retained for debugging, but it must be stored outside the accepted training-data index and labeled with reasons such as:

- underresolved current sheet;
- time-step convergence failure;
- energy-balance failure;
- incomplete event horizon;
- topology-tracking ambiguity;
- parity failure; or
- nonfinite state.

Module 3 may ingest only accepted cases by default. A trajectory that fails the original numerical criteria does not become usable merely because a neural operator consumes full arrays.

13 Python and MATLAB implementation handoff

13.1 Parallel scientific contract

Both implementations must use the same:

- equations and signs;

- grid and endpoint convention;
- initial-condition formulas;
- FFT normalization and wave-number definitions;
- dealiasing mask;
- RK4 algorithm;
- zero-mode policy;
- diagnostic definitions;
- output times and array-axis contract;
- raw full-field HDF5 handoff and operator-interface version; and
- verification tolerances.

They need not have identical internal function names, but their scientific operations must be traceable one-to-one.

13.2 Recommended components

Component	Responsibility
Configuration loader	Parse values, reject invalid grids or coefficients, and freeze defaults.
Spectral grid	Build physical coordinates, FFT wave numbers, k^2, zero-mode mask, and two-thirds mask.
Initial-condition generator	Evaluate $\psi_0, U_0, J_0, \omega_0$ and analytic checks.
Poisson solver	Reconstruct zero-mean U from ω.
Bracket operator	Evaluate dealiased $[f, g]$ with shared derivatives.
Right-hand side	Assemble Eqs. (5.1)–(5.2).
RK4 stepper	Advance the stage-consistent Fourier state.
Diagnostics	Compute fields, balances, topology, spectra, and event markers.
Case writer	Write atomic HDF5 numerical arrays and manifest metadata without rendering or lossy field conversion.
Verification suite	Run exact, manufactured, convergence, and parity tests.

13.3 One-right-hand-side parity checkpoint

Before long trajectories are compared, export the initial Fourier state and one evaluated right-hand side from both languages. This localizes differences:

- disagreement in $\widehat{J}$ indicates Laplacian or wave-number error;
- disagreement in $\widehat{U}$ indicates Poisson inversion or zero-mode error;
- disagreement only in brackets indicates derivative, transpose, or dealiasing error;
- agreement in the right-hand side but disagreement after one step indicates RK4 staging error; and
- agreement after one step but later drift requires time-history and transform-order analysis.

13.4 Full-field interface checks

In addition to the PDE-solver verification, the operator-ready handoff must pass the following inexpensive checks:

1. **Initial-target identity.** At $t_0 = 0$, the target fields read from storage must equal the stored input fields ψ_0, U_0 to the accepted storage precision.
2. **Tensor round trip.** Reading and writing a field tensor with no intentional transform must reproduce its values, shape, and axis order.
3. **Grid registration.** A known analytic Fourier mode or critical point must appear at the same physical coordinates after Python–MATLAB serialization and readback.
4. **Parameter-field consistency.** For the frozen analytic family, the amplitude inferred from $U_0 = A_0(\cos x - \cos y)$ must agree with stored A_0 within tolerance.
5. **No-rendering provenance.** Learning tensors must trace directly to the accepted HDF5 arrays; visualization products are outside the regression path.

These checks do not validate the neural operator. They verify that Module 2 hands the correct numerical fields to Module 3 without silent transposition, resampling, or image conversion.

13.5 Acceptance matrix

Test	Frequency	Acceptance meaning
Fourier derivative and Poisson tests	Every software revision	Primitive operators are correct.
Bracket identities	Every software revision	Nonlinear operator and mask are consistent.
Exact resistive decay	Every software revision	Diffusion sign and RK4 order are correct.
Manufactured nonlinear solution	Major solver revision	All terms converge to a known solution.
Energy, means, divergence, and symmetry	Every case	The realized trajectory respects structural identities.
Temporal refinement	Each new numerical regime	Time step is accurate for that regime.
Spatial refinement and spectra	Each new resolution regime	Current sheets and derived fields are resolved.
Python–MATLAB parity	Baseline and representative extremes	Independent implementations agree.
HDF5/tensor round trip	Every writer revision	Stored arrays preserve values, shape, axes, precision, and metadata.
Initial-target identity	Every schema/interface revision	The $t = 0$ operator target reproduces the input fields exactly within storage tolerance.

14 Common failure modes

14.1 A duplicated periodic endpoint

Using both 0 and 2π creates $N + 1$ samples for N Fourier intervals and breaks the assumed FFT grid. Use a half-open grid.

14.2 An incorrect FFT wave-number array

The upper half of FFT indices represents negative frequencies. Treating every index as positive gives incorrect derivatives despite visually periodic fields.

14.3 A zero-mode division

Replacing $k^2 = 0$ by a small arbitrary number contaminates the stream-function gauge. Set $\widehat{U}_{00} = 0$ explicitly.

14.4 A sign copied from another convention

Many papers define velocity, vorticity, current, or brackets with the opposite sign. Translate the entire set together. Copying only one equation changes the physics.

14.5 No dealiasing or inconsistent dealiasing

Aliased nonlinear products can inject false energy and small structures. Applying different masks in Python and MATLAB invalidates parity.

14.6 Stale stage fields

Reusing U or J from the beginning of an RK4 step lowers the method's order and changes nonlinear dynamics.

14.7 An apparently stable but inaccurate time step

No blow-up does not imply correct peak timing or amplitude. Require temporal refinement.

14.8 Energy-only verification

Global integrals can hide local phase, topology, and derivative errors. Include fields, current sheets, reconnection, spectra, and exact solutions.

14.9 A current sheet thinner than the grid

An unresolved sheet can show a plausible single-cell peak. Require width, spectral-tail, and refinement evidence.

14.10 Finite-differencing sparse reconnection data

Sparse output can produce a noisy or biased rate. Prefer the PDE value of ψ_t at tracked critical points and use saved-value differences only as a cross-check.

14.11 A fixed X-point assumption after symmetry breaks

The central coordinate is a useful check, not a universal tracker. Locate and classify critical points.

14.12 Treating every finite run as accepted

Execution success, verification success, numerical convergence, and dataset acceptance are different flags.

14.13 Treating numerical noise as physical plasmoids

Secondary islands can be seeded by underresolution or algorithmic noise. Confirm them under spatial and temporal refinement and controlled perturbation amplitude.

14.14 Generating the sweep before converging the baseline

This creates a large dataset whose errors are unknown. Converge one representative nonlinear case first.

14.15 Treating rendered images as field data

A contour or field-line image can look physically convincing while discarding exact scalar values, signs, coordinates, and precision. Feeding rendered plots to the surrogate changes the scientific problem into computer vision and breaks the numerical field contract. Use the raw HDF5 arrays for learning and reserve images for human visualization.

14.16 Counting time-conditioned pairs as independent cases

Many targets can be extracted from one trajectory, but they share the same initial state and evolution. Randomly splitting those pairs across train and test sets creates trajectory leakage and exaggerates dataset breadth. Split complete accepted cases first in Module 3.

14.17 Claiming M3D-C1 fidelity

The solver uses a periodic Cartesian two-field reduced model. It does not reproduce M3D-C1 geometry, closures, finite elements, or additional evolved fields. Using full-field arrays or formatting their schema to resemble a larger simulation output does not change that claim boundary.

15 Module 2 computational contract

Frozen Module 2 convention: mathematical problem

$$\Omega = [0, 2\pi)^2, \qquad \partial_t \psi + [U, \psi] = \eta \nabla_\perp^2 \psi,$$
$$\partial_t \omega + [U, \omega] = [J, \psi] + \nu \nabla_\perp^2 \omega,$$
$$J = -\nabla_\perp^2 \psi, \qquad \omega = \nabla_\perp^2 U, \qquad \langle U \rangle = \langle \psi \rangle = 0,$$
$$\psi_0 = \sin x \sin y, \qquad U_0 = A_0(\cos x - \cos y).$$

Frozen Module 2 convention: numerical method

Advance $(\widehat{\psi}, \widehat{\omega})$ with a doubly periodic Fourier pseudo-spectral method, rectangular two-thirds dealiasing, zero-mode-safe Poisson inversion, and classical fixed-step RK4. Reconstruct U and J at every RK stage.

Frozen Module 2 convention: baseline

$$(\eta, \nu, A_0) = (10^{-3}, 10^{-3}, 0.1), \quad N_x = N_y = 128, \quad \Delta t = 10^{-3}, \quad \Delta t_{\mathrm{save}} = 0.05.$$

Use $T = 5$ for the first pilot. Extend the accepted baseline nominally to $T = 8$ so that it includes the first resolved reconnection/current peak and a post-peak interval.

Frozen Module 2 convention: full-field operator handoff

Module 2 publishes complete accepted HDF5 trajectories containing raw numerical arrays. The primary downstream view is

$$(\psi_0, U_0, \eta, \nu, t_n) \longmapsto (\psi_n, U_n).$$

The fields are two-dimensional scalar arrays, not rendered images. A_0 remains required case metadata but is already encoded in U_0 for the frozen initial-condition family. Module 3 performs case splitting, time selection, scaling, and architecture-specific tensor packing.

Frozen Module 2 convention: acceptance

No trajectory becomes training data until numerical-operator tests, exact or manufactured verification, energy and structural identities, temporal refinement, spatial refinement, spectral resolution, topology diagnostics, and Python–MATLAB parity have passed under recorded tolerances.

16 Summary and bridge to Module 3

Module 2 converts the reduced-MHD model into an auditable numerical reference. The solver evolves magnetic flux and vorticity, reconstructs stream function and current, enforces periodicity through Fourier representation, removes quadratic aliasing through two-thirds truncation, and advances the semi-discrete system with classical RK4.

The selected initial state is physically and numerically useful. Its magnetic part is an exact Laplacian eigenfunction satisfying $J_0 = 2\psi_0$, so the unperturbed field is an ideal equilibrium and has an exact finite-resistivity decay. The stream function $U_0 = A_0(\cos x - \cos y)$ drives two parallel negative-current islands toward the central X-point, where a current sheet and resistive reconnection develop.

Verification proceeds from exact operators to nonlinear manufactured solutions, balance laws, refinement studies, spectral tails, topology tracking, and independent-language parity. This hierarchy is essential because no single diagnostic can certify a nonlinear reconnection trajectory.

The revised surrogate interface does not change the reduced-MHD solver. It changes the downstream interpretation of each accepted trajectory: the complete initial arrays (ψ_0, U_0), together with (η, ν, t), condition a prediction of the complete later arrays (ψ_t, U_t). Numerical arrays remain authoritative; rendered images are visualization only. For the present benchmark, A_0 is retained as case metadata but is already encoded in U_0, and the fixed-shape initial-condition family limits the strength of any claim about arbitrary-field generalization.

Module 3 receives only accepted complete cases. Its job is to preserve axes and provenance, assign whole trajectories to train/validation/test splits before extracting target times, and transform the raw numerical fields into a versioned full-field operator dataset without changing their physical meaning.

A Detailed sign audit for the initial condition

From $\psi_0 = \sin x \sin y$,

$$\psi_{0,x} = \cos x \sin y, \qquad \psi_{0,y} = \sin x \cos y, \tag{A.1}$$

$$\psi_{0,xx} = -\sin x \sin y, \qquad \psi_{0,yy} = -\sin x \sin y. \tag{A.2}$$

Therefore

$$\nabla_\perp^2 \psi_0 = -2\psi_0, \qquad J_0 = -\nabla_\perp^2 \psi_0 = 2\psi_0. \tag{A.3}$$

From $U_0 = A_0(\cos x - \cos y)$,

$$U_{0,x} = -A_0 \sin x, \qquad U_{0,y} = +A_0 \sin y, \tag{A.4}$$

$$U_{0,xx} = -A_0 \cos x, \qquad U_{0,yy} = +A_0 \cos y. \tag{A.5}$$

Thus

$$\omega_0 = \nabla_\perp^2 U_0 = -U_0, \qquad \boldsymbol{v}_{\perp,0} = (-A_0 \sin y, -A_0 \sin x). \tag{A.6}$$

B Parseval normalization

With the unnormalized forward discrete transform and inverse factor $1/(N_x N_y)$,

$$\sum_{j,\ell} |f_{j\ell}|^2 = \frac{1}{N_x N_y} \sum_{m,n} |\widehat{f}_{mn}|^2. \tag{B.1}$$

The physical-space quadrature is

$$\int_\Omega |f|^2 \, \mathrm{d}A \approx \Delta x \Delta y \sum_{j,\ell} |f_{j\ell}|^2. \tag{B.2}$$

Therefore a Fourier-space diagnostic must include the transform and quadrature factors consistently. Cross-check every spectral energy against its physical-space derivative form.

C Observed-order formulas

If $e(h) = Ch^p + o(h^p)$, then

$$p \approx \log_2 \left(\frac{e(h)}{e(h/2)} \right). \tag{C.1}$$

When exact truth is unavailable, three numerical solutions give

$$p_{\mathrm{obs}} \approx \log_2 \frac{\left\| q_h - q_{h/2} \right\|}{\left\| q_{h/2} - q_{h/4} \right\|}. \tag{C.2}$$

This estimate is meaningful only when all three levels lie in the same asymptotic regime and are compared at the same physical state.

D Symbol table

Symbol	Meaning	Status
x, y	Periodic Cartesian coordinates	dimensionless
t, T	Time and final time	dimensionless
L_x, L_y	Domain periods	2π in baseline
N_x, N_y	Grid points in each direction	integers
$\Delta x, \Delta y$	Uniform grid spacings	$L_x/N_x, L_y/N_y$
Δt	Integrator step size	dimensionless
Δt_{save}	Saved-snapshot interval	dimensionless
Ω	Periodic domain $[0, 2\pi)^2$	dimensionless area
ψ	Magnetic-flux function	primary stored field
U	Flow stream function	primary stored field
J	Current variable $-\nabla_\perp^2 \psi$	derived field
ω	Vorticity $\nabla_\perp^2 U$	prognostic and stored field
$\boldsymbol{B}_\perp$	In-plane magnetic field $(\psi_y, -\psi_x)$	derived vector
$\boldsymbol{v}_\perp$	In-plane velocity $(-U_y, U_x)$	derived vector
η	Magnetic diffusivity	case parameter
ν	Kinematic viscosity	case parameter
A_0	Initial stream-function perturbation amplitude	case metadata; encoded in U_0 for the frozen operator input
Ψ_{eq}	Fixed equilibrium flux amplitude	1 in version 1
S	Lundquist number under reference normalization	$1/\eta$
Re	Reynolds number under reference normalization	$1/\nu$
Pm	Magnetic Prandtl number	ν/η
$[f, g]$	Poisson bracket $f_x g_y - f_y g_x$	nonlinear operator
$\widehat{f}$	Discrete Fourier coefficient of f	complex numerical value
k_x, k_y	Fourier wave numbers	integer for 2π period
k^2	$k_x^2 + k_y^2$	nonnegative
$\mathcal{P}_{2/3}$	Rectangular two-thirds spectral projection	dealiasing operator
E_B, E_K, E	Magnetic, kinetic, and total energy	diagnostics
$\mathcal{D}_\eta, \mathcal{D}_\nu$	Ohmic and viscous dissipation rates	diagnostics

Symbol	Meaning	Status
δ_E	Integrated normalized energy-balance defect	verification metric
$\Delta\psi_{OX}$	O-to-X flux contrast	topology diagnostic
$\Phi_{\rm rec}$	Signed reconnected flux	topology diagnostic
$\mathcal{R}_{\rm rec}$	Signed reconnection rate	topology diagnostic
$J_{\max}$	Maximum absolute current	sheet diagnostic
H_ψ	Hessian matrix of ψ	critical-point classifier
$p_{\rm obs}$	Observed convergence order	numerical metric
$\mathcal{G}$	Full-field solution-operator map in Eq. (1.4)	downstream learning interface
X_n, Y_n	Initial/context input and full-field target at saved time t_n	operator-learning pair
$\epsilon_{\rm mach}$	Machine roundoff scale	numerical constant

E Acronym and terminology table

Term	Meaning
2D	Two-dimensional
BC	Boundary condition
CFL	Courant–Friedrichs–Lewy; a stability restriction linking time step, grid spacing, and propagation speed
DFT	Discrete Fourier transform
FFT	Fast Fourier transform, an efficient DFT algorithm
HDF5	Hierarchical Data Format version 5
IC	Initial condition
M3D-C1	Extended-MHD simulation framework using high-order C^1 finite elements
MHD	Magnetohydrodynamics
MMS	Method of manufactured solutions
Neural operator	Learned mapping whose input and/or output is a complete function or discretized field rather than only one point value
Full-field pair	One operator input/target pair built from complete numerical scalar arrays on the accepted grid
Raw numerical field	Array of scalar field values with coordinates, sign, amplitude, and numerical precision preserved; not a rendered image
O-point	Elliptic magnetic critical point at an island center
PDE	Partial differential equation
PINN	Physics-informed neural network

Term	Meaning
RK4	Classical fourth-order Runge–Kutta method
RMHD	Reduced magnetohydrodynamics; the precise reduction must be stated
X-point	Hyperbolic magnetic critical point on a separatrix
Aliasing	Folding of unresolved product modes into incorrect resolved modes
Collocation grid	Physical points at which pseudo-spectral fields and nonlinear products are evaluated
Convergence	Approach toward a stable numerical answer under refinement
Dealiasing	Removal or exact treatment of modes that would corrupt nonlinear products
Fourier mode	Periodic complex exponential basis function with a specified wave vector
Hermitian symmetry	Coefficient symmetry required for a Fourier series to represent a real field
Island coalescence	Attraction, current-sheet formation, reconnection, and merger of parallel-current magnetic islands
Method of lines	Spatial discretization of a PDE followed by time integration of the resulting ODE system
Pseudo-spectral	Spectral differentiation with nonlinear products evaluated on a collocation grid
Reconnection	Non-ideal change of magnetic-field-line connectivity
Self-convergence	Comparison of nested numerical solutions when no exact nonlinear solution is available
Spectral tail	Energy or amplitude in the highest retained Fourier modes
Verification	Evidence that the numerical implementation solves the stated equations correctly
Validation	Evidence that the equations adequately represent an external physical target

F Concept questions and answer sketches

Questions

Q1. Why are ψ and ω convenient prognostic variables even though the surrogate predicts ψ and U?

Q2. Why must the grid omit the point $x = 2\pi$?

Q3. Why is $\widehat{U}_{00}$ not determined by ω?

Q4. Show that $J_0 = 2\psi_0$ for the island array.

Q5. Why does $[J_0, \psi_0] = 0$?

Q6. What physical role does $A_0(\cos x - \cos y)$ play?

Q7. Which two islands move toward the central X-point for $A_0 > 0$?

Q8. Why is the $A_0 = 0$ finite-resistivity solution useful for verification?

Q9. What is aliasing, and why must dealiasing occur inside every right-hand-side evaluation?

Q10. Why must U and J be recomputed at every RK4 stage?

Q11. Why does a stable run still require time-step refinement?

Q12. Why does good energy behavior not prove that reconnection is correct?

Q13. How can a current sheet be finite but unresolved?

Q14. Why is $J_{\max}$ more resolution-sensitive than ψ?

Q15. Why must O-points and X-points be tracked rather than always sampled at their initial coordinates?

Q16. Why is the PDE value of ψ_t useful for reconnection rate?

Q17. What is the difference between temporal and spatial convergence studies?

Q18. Why should the parameter sweep begin only after the baseline converges?

Q19. Why is optional noise a physical case parameter rather than a harmless coding detail?

Q20. What does Python–MATLAB agreement establish, and what does it not establish?

Q21. Why can this module support a reduced-MHD surrogate but not an M3D-C1 surrogate claim?

Q22. What is the revised full-field operator map produced from an accepted trajectory?

Q23. Why are contour plots or magnetic-field-line images not valid substitutes for the numerical fields?

Q24. Why does supplying the complete U_0 field make a duplicate A_0 neural input unnecessary for the frozen family, while A_0 still remains metadata?

Q25. Why does the full-field input representation not yet demonstrate generalization to arbitrary initial conditions?

Q26. Why must train/validation/test splitting occur by complete trajectory before extracting multiple target times?

Answer sketches

A1. The vorticity equation supplies ω_t directly; U is recovered by a diagonal Fourier Poisson inversion. The stored physical fields remain (ψ, U).

A2. The endpoints are the same periodic location. Duplicating one violates the N-point Fourier collocation convention.

A3. A spatial constant in U has zero Laplacian and no physical velocity. The gauge sets that constant.

A4. Each second derivative contributes $-\sin x \sin y$, so $-\nabla_\perp^2 \psi_0 = 2\psi_0$.

A5. J_0 is a scalar multiple of ψ_0, and the bracket of any field with itself is zero.

A6. It is a symmetric initial flow that selects and accelerates the coalescence mode.

A7. The negative-current islands at $(\pi/2, 3\pi/2)$ and $(3\pi/2, \pi/2)$ move toward (π, π).

A8. It gives an exact analytic field, decay rate, derived current, and zero flow.

A9. Aliasing misidentifies an unresolved product mode as a lower resolved mode. Because it changes the right-hand side, output-only filtering is too late.

A10. Every stage is a different state at which the nonlinear right-hand side must be evaluated consistently.

A11. Stability prevents blow-up; it does not bound phase, peak, or event-time error tightly enough.

A12. Energy is a global scalar. Local topology, signs, and current-sheet structure may still be wrong.

A13. A one- or two-cell sheet has finite samples but lacks enough points to represent its width and peak.

A14. J contains two spatial derivatives, which amplify high-wave-number error.

A15. Critical points move during coalescence and may be created or destroyed.

A16. At a critical point, the correction from point motion vanishes because $\nabla\psi = 0$.

A17. Temporal refinement holds space fixed and changes Δt; spatial refinement holds time error small and changes N.

A18. Otherwise every generated trajectory inherits an unknown common discretization error.

A19. Noise can seed physical instabilities and change the reconnection regime, especially at high Lundquist number.

A20. It reduces the probability of a language-specific implementation error. Two matching codes can still share the same mistaken specification.

A21. It verifies only the stated two-field periodic Cartesian model, not M3D-C1's full equations, geometry, or numerical method.

A22. It is $(\psi_0, U_0, \eta, \nu, t_n) \mapsto (\psi_n, U_n)$, where each ψ and U is the complete numerical field on the accepted two-dimensional grid.

A23. Rendering is a lossy, presentation-dependent transformation that can discard exact scalar values, signs, coordinates, and precision. The operator should learn from the original arrays.

A24. In version 1, $U_0 = A_0(\cos x - \cos y)$, so the entire array already determines its amplitude. A_0 is still stored because it is part of the physical case definition and is useful for provenance and stratified analysis.

A25. The version-1 initial fields vary only within a fixed-shape family, chiefly through A_0. A network seeing full arrays has a richer representation, but the data still do not contain arbitrary initial-field shapes.

A26. Different times from one trajectory are dependent samples of the same physical evolution. Splitting them independently leaks trajectory information and can falsely inflate unseen-case performance.

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